FULL-STACK WEB APPLICATION

Employee Management System

A role-based platform for managing employees, leaves, helpdesk tickets — built with Angular & Spring Boot.

Angular 17+ Spring Boot Java 21 REST API JWT Auth

Role-Based Access

ABOUT THE SYSTEM

What is EMS?

Employee Management

Manage all employee records — add, edit, delete, search, and paginate through the full directory.

Leave Management

Employees apply for leave. Managers review and approve or reject requests in real time.

Helpdesk Tickets

Employees raise support tickets by category. Full history with status tracking.

Secure Auth

JWT-based login with role-aware routing. Guards protect every page from unauthorised access.

Chat Assistant

Floating AI chat widget with quick actions — apply leave, create ticket, navigate by role.

Role Dashboards

Every role gets a personalised dashboard — profile cards, team metrics, and direct shortcuts.

BUILT WITH

Technology Stack

🖥️ FRONTEND

- Angular 17+
- TypeScript
- Angular Material
- RxJS
- Angular Router
- HTTP Interceptors
- Route Guards
- SCSS / Material Theme

⚙️ BACKEND & SECURITY

- Spring Boot
- Java 21
- Gradle
- REST API
- JWT Authentication
- Role-Based Access
- SMTP Email
- Password Hashing

ACCESS CONTROL

4 User Roles in the System

Admin

Full system control — CRUD, password resets, all features.

HR

Manage all employee records, add new staff, update details.

Manager

View team directory, approve or reject leave requests.

Employee

Self-service — profile, leave, tickets, and chat assistant.

Feature	Admin	HR	Manager	Employee
Employee CRUD	✅ Full	✅ Full	👁 View only	—
Leave Management	—	—	✅ Approve/Reject	✅ Apply/View
Helpdesk Tickets	—	—	—	✅ Create/View
Reset Password	✅ Any user	—	—	—
Chat Assistant	✅	✅	✅	✅

ROLE DEEP-DIVE

What can the Admin do?

Admin

The Admin has the highest level of access. They control the entire employee directory, can reset passwords for any user, and use the chat assistant for fast navigation.

1 **Login & Redirect**
Login with admin credentials → system redirects to `/admin` dashboard.

2 **View Employee Directory**
See all employees with name, role, department, salary, email and joining date on card layout.

3 **Add a New Employee**
Click Add → fill first name, last name, email, role, department, salary, joining date → submit.

4 **Edit / Delete Employee**
Click Edit on any card to update details. Click Delete → confirm dialog → employee removed.

5 **Reset Any Password**
Enter employee email + new password → submit. Useful when an employee is locked out.

6 **Use Chat Assistant**
Click chat widget → use quick actions or type commands to navigate the system instantly.

ROLE DEEP-DIVE

What can HR do?

HR

HR manages the full employee lifecycle — onboarding new staff, updating records, and maintaining the directory. They have the same CRUD capabilities as Admin, minus password resets.

1 **Login & Redirect**
Login → redirected to `/hr` dashboard with the full employee directory view.

2 **Search & Filter Employees**
Use real-time debounced search by name, email, role, department, salary, or joining date.

3 **Onboard New Employee**
Click Add Employee → complete the form → system POSTs to API → snackbar confirms success.

4 **Update Employee Records**
Click Edit on any employee card → modify fields → save. Changes reflected immediately in the directory.

5 **Remove an Employee**
Click Delete → confirmation dialog appears → confirm → employee removed with snackbar feedback.

6 **Paginate & View Metrics**
Page through 6 / 12 / 24 / All employees. View total staff count and department metrics at a glance.

ROLE DEEP-DIVE

What can the Manager do?

Manager

Managers focus on their team — they can view the employee directory in read-only mode and are the decision-makers for all leave requests from their team members.

1 **Login & Redirect**
Login → redirected to `/manager` dashboard with leave queue and team directory.

2 **View Team Directory**
Browse all employees — name, role, department, email visible. No add/edit/delete options (read-only).

3 **View Pending Leaves**
See all leave requests submitted by employees waiting for a decision — type, dates, reason shown.

4 **Approve a Leave**
Click Approve on a pending request → API call to `POST /approve` → status updates instantly.

5 **Reject a Leave**
Click Reject on a pending request → API call to `POST /reject` → request marked rejected.

6 **Use Chat Assistant**
Use chat to quickly navigate to the leave queue, employee list, or get answers about the system.

ROLE DEEP-DIVE

What can the Employee do?

Employee

Employees have a self-service workspace — they manage their own profile, apply for leave, raise helpdesk tickets, and use the chat assistant for fast navigation.

1 **Login & Dashboard**
Login → redirected to `/employee` showing profile card, shortcuts, and company feed.

2 **View Own Profile**
See personal details — name, ID, email, department, role, salary, and joining date on a profile card.

3 **Apply for Leave**
Click Apply Leave → choose leave type, start & end dates, add reason → submit → pending status shown.

4 **View Leave History**
See all past and current leave requests with status — Pending / Approved / Rejected.

5 **Raise a Helpdesk Ticket**
Click Create Ticket → select category → fill subject & description → submit → ticket tracked in history.

6 **Chat Assistant**
Use quick actions — Apply Leave, Create Ticket, Check Payslip, Reset Password — all via chat widget.

BUILT-IN FEATURE

Chat Assistant

Every user gets access to the floating **Chat Assistant widget** — available on all pages after login. It understands intent and routes users to the right screen instantly.

Apply Leave

Navigates employee to the leave section of their dashboard.

Create Ticket

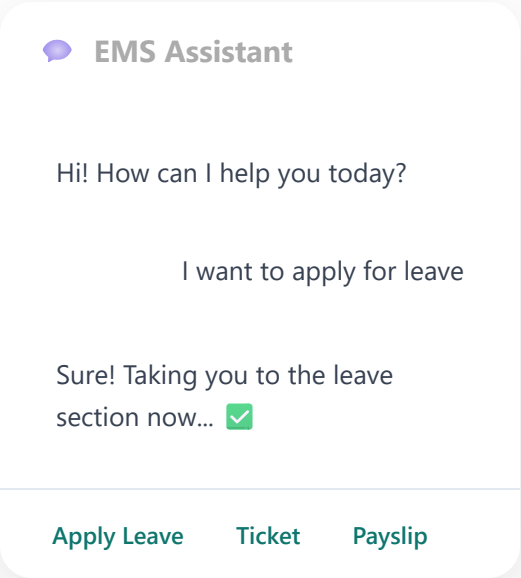
Takes employee directly to the helpdesk ticket form.

Check Payslip

Shows a response message (payslip feature placeholder).

Reset Password

Admin-only action — routes to the password reset form.



TECHNICAL DESIGN

System Architecture

FRONTEND — ANGULAR

🔐 Auth & Route Guards

📄 Role-Based Components

🔄 HTTP Interceptor (JWT)

💬 Chat Widget

⟷

BACKEND — SPRING BOOT / JAVA 21

🔑 /api/auth/*

👥 /api/employees/*

📅 /api/employee/leaves/*

📧 SMTP Email Service

⟷

SECURITY LAYER

🪙 JWT Token Issuing

🔒 Role Verification

🛡️ Token Expiry & Refresh

🔐 Password Hashing

Angular runs on **localhost:4200** · Spring Boot API runs on **localhost:8080** · JWT stored in **localStorage** per session

HOW THE APP STAYS SECURE

Security & Authentication

1 **JWT Token on Login**
After successful login the server returns a signed JWT token which is stored in localStorage.

2 **HTTP Interceptor**
Every outgoing API request automatically has the JWT token attached in the Authorization header.

3 **Role-Based Route Guard**
Angular guards check the stored role before activating any route — wrong role = redirect to login.

4 **Backend Role Verification**
Every API endpoint on the server checks the JWT role claim — a Manager cannot call admin endpoints.

5 **Logout Clears Session**
Logout removes token, role, employeeId, and email from localStorage — session fully wiped.

6 **Password Hashing**
Passwords are never stored in plain text — hashed on the server before saving to the database.

GETTING STARTED

How to Run the Application

1 **Install Prerequisites**
Node.js v18+ · Angular CLI · Git

2 **Start the Backend (Spring Boot)**
`cd memployee` → `./gradlew bootRun` → runs on `localhost:8080`

3 **Install Frontend**
`cd FIRSTAPP` → `npm install`

4 **Start Angular Server**
`npm start` or `ng serve`

5 **Open in Browser**
Visit `http://localhost:4200` → login page appears.

6 **Login with Your Role**
Enter credentials → system auto-redirects to your role's dashboard.

⚙️ Backend: **localhost:8080**

🌐 Frontend: **localhost:4200**

🔑 Login page: **/login**

WHAT MAKES IT STAND OUT

Key Feature Highlights

Smart Search

Debounced real-time search across name, email, role, department & salary.

Route Guards

Role-protected routes — access the wrong page and you're redirected instantly.

Chat Assistant

Intent-aware chat with quick actions and role-specific navigation commands.

Live Feedback

Snackbar alerts for every action — add, edit, delete, approve, reject.

Pagination

Directory paginates at 6 / 12 / 24 / All — handles large staff lists cleanly.

Modern UI

Gradient backgrounds, hero cards, animated particles, and Material Design.

Real-time Updates

All changes reflect immediately — no page reloads needed, pure Angular reactivity.

JWT Security

Token auto-attached to every request via HTTP interceptor — fully stateless auth.

Thank You

Employee Management System

Built with Angular · Spring Boot · Java 21 · Gradle · JWT

🛡️ Admin 📋 HR 👔 Manager 👷 Employee

Use ← → arrow keys to navigate · Click Save PDF to export